GLYCOTWIN SYSTEM ARCHITECTURE · TEAM CLASSY LEOPARDS (GOA INSTITUTE OF MANAGEMENT, SANQUELIM)

Multimodal Ingestion, Feature Kinetics, AI Scoring & Decision Pipeline

LAYER 1

Multimodal Ingestion

- **Static Clinical EHR:** Age, Gender, South-Asian BMI, Duration
- **Baseline Biomarkers:** HbA1c (5.7–12.5%), eGFR, Fasting Glucose
- **Pharmacotherapy:** Metformin, SU, SGLT2i, Insulin
- **Dynamic CGM Telemetry:** 15-min Glucose Readings
- **Optical PPG:** Heart Rate (BPM), HRV Stress (ms)
- **Inertial Accelerometer:** 15-min Steps, Sleep State

LAYER 2

Time-Series Kinetics

- **Rolling Derivatives:** Glucose Velocity dG/dt (15m, 30m)
- **Glycemic Volatility:** 1-hour Rolling Standard Deviation
- **Metabolic Accumulator:** 1h & 2h Step Integration
- **Nutrition Kinetics:** 2-hour Carbohydrate Absorption
- **Pharmacodynamics:** 2-hour Bolus Insulin Integration
- **Synchronous Fusion Engine:** Patient Window Alignment

LAYER 3

Predictive AI Suite

- **Event Classifier:** 2h Impending Hypo/Hyper Risk
- **Trajectory Regressor:** Continuous Glucose in mg/dL
- **Cost-Sensitive Balancing:** Zero Missed Hypo Nadirs
- **Native Missing Values:** Zero Imputation Distortion
- **Calibrated Probability:** Exact Confidence Score (%)
- **Sub-5ms Inference:** Client-Side Edge Execution

LAYER 4

Clinical Decision

- **Early Warning Horizon:** 120-Minute Alert Window
- **Rescue Protocols:** 15–20g Fast-Acting Carb Intake
- **What-If Sandbox:** Virtual Food & Insulin Titration
- **5-Step Sensor Audit:** Continuity & Bound Checks
- **Population Analytics:** ADA 7.0% Triage Registry
- **Global Edge CDN:** Live at glycotwin-app.web.app